Problem 1 : Smarr formula

$$M_H = -\frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\mu (\partial t)_\nu \, dS^{\mu\nu} = -\frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\mu (\partial t)_\nu \sqrt{-g}\, dx^\mu dx^\nu$$

在视界上 $dt = dr = 0$; $\Delta = 0$, $r = r_+$ 从而:

$$M_H = -\frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\phi (\partial t)_\phi \sqrt{-g}\, d\phi d\phi - \frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\theta (\partial t)_\theta \sqrt{-g}\, d\theta d\theta - \frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\theta (\partial t)_\phi \sqrt{-g}\, d\theta d\phi$$

$$- \frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\phi (\partial t)_\theta \sqrt{-g}\, d\phi d\theta$$

$\because (\partial t)^\mu = (1,0,0,0)$

$$\therefore (\partial t)_\mu |_{\mathcal{H}} = g_{\mu\nu} (\partial t)^\nu |_{\mathcal{H}} = g_{00} (\partial t)^0 + g_{40} (\partial t)^0$$

$$= \left(\frac{a^2 \sin^2\theta}{r_+^2 + a^2\cos^2\theta}, 0, 0, \frac{-a\sin^2\theta (r_+^2 + a^2)}{r_+^2 + a^2\cos^2\theta} \right) \Rightarrow (\partial t)_\theta = 0$$

从而 $M_H = -\frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\phi (\partial t)_\phi \sqrt{-g}\, d\phi d\phi - \frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\theta (\partial t)_\phi \sqrt{-g}\, d\theta d\phi$

其中 $\nabla_\phi (\partial t)_\phi = \partial_\phi (\partial t)_\phi - \Gamma^t_{\phi\phi} (\partial t)_t - \Gamma^\phi_{\phi\phi} (\partial t)_\phi = 0$

$$\nabla_\theta (\partial t)_\phi = \partial_\theta (\partial t)_\phi - \Gamma^t_{\theta\phi} (\partial t)_t - \Gamma^\phi_{\theta\phi} (\partial t)_\phi$$

$$= \frac{d}{d\theta}\left[\frac{-a\sin^2\theta (r_+^2 + a^2)}{r_+^2 + a^2\cos^2\theta} \right] - \left(-\frac{4a^3(-2Mr_+)\cos\theta\sin^3\theta}{(a^2 + 2r_+^2 + a^2\cos 2\theta)^2} \right) \frac{a^2\sin^2\theta}{r_+^2 + a^2\cos^2\theta} - \cdots$$

$$J_H = -\frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\mu (\partial\phi)_\nu \, dS^{\mu\nu} = -\frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\mu (\partial\phi)_\nu \sqrt{-g}\, dx^\mu dx^\nu$$

$$= -\frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\phi (\partial\phi)_\phi \sqrt{-g}\, d\phi d\phi - \frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\theta (\partial\phi)_\theta \sqrt{-g}\, d\theta d\theta - \frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\theta (\partial\phi)_\phi \sqrt{-g}\, d\theta d\phi$$

$$- \frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\phi (\partial\phi)_\theta \sqrt{-g}\, d\phi d\theta$$

$\because (\partial\phi)^\mu = (0,0,0,1)$

$$\therefore (\partial\phi)_\mu |_{\mathcal{H}} = g_{\mu\nu} (\partial\phi)^\nu |_{\mathcal{H}} = g_{04} (\partial\phi)^4 + g_{44} (\partial\phi)^4$$

$$= \left(\frac{-a\sin^2\theta (r_+^2 + a^2)}{r_+^2 + a^2\cos^2\theta}, 0, 0, \frac{(r_+^2 + a^2)^2 \sin^2\theta}{r_+^2 + a^2\cos^4\theta} \right) \Rightarrow (\partial\phi)_\theta = 0$$

$$\therefore J_H = -\frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\phi (\partial\phi)_\phi \sqrt{-g}\, d\phi d\phi - \frac{1}{8\pi}\oint_{\mathcal{H}} \nabla_\theta (\partial\phi)_\phi \sqrt{-g}\, d\theta d\phi$$

其中: $\nabla_\phi (\partial\phi)_\phi = \partial_\phi (\partial\phi)_\phi - \Gamma^t_{\phi\phi} (\partial\phi)_t - \Gamma^\phi_{\phi\phi} (\partial\phi)_\phi = 0$

$$\nabla_\theta (\partial\phi)_\phi = \partial_\theta (\partial\phi)_\phi - \Gamma^t_{\phi\phi} (\partial\phi)_\phi - \Gamma^\phi_{\phi\phi} (\partial\phi)_\phi =$$

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$$\langle \phi_I', \phi_J' \rangle = \sum_k \sum_l \langle \alpha_{Ik}\phi_k + \beta_{Ik}\phi_k^*, \alpha_{Jl}\phi_l + \beta_{Jl}\phi_l^* \rangle$$
$$= \sum_k \sum_l \alpha_{Ik}^*\alpha_{Jl}\langle \phi_k, \phi_l \rangle + \beta_{Ik}^*\beta_{Jl}\langle \phi_k^*, \phi_l^* \rangle$$
$$= \sum_k \sum_l \delta_{kl}(\alpha_{Ik}^*\alpha_{Jl} - \beta_{Ik}^*\beta_{Jl})$$
$$= \sum_k \alpha_{Ik}^*\alpha_{Jk} - \beta_{Ik}^*\beta_{Jk}$$
$$= (\alpha\alpha^\dagger - \beta\beta^\dagger)_{IJ} = \delta_{IJ} \quad \Rightarrow \alpha\alpha^\dagger - \beta\beta^\dagger = 1$$

$$\langle \phi_I', \phi_J'^* \rangle = \sum_k \sum_l \langle \alpha_{Ik}\phi_k + \beta_{Ik}\phi_k^*, \beta_{Jl}^*\phi_l + \alpha_{Jl}^*\phi_l^* \rangle$$
$$= \sum_k \sum_l \alpha_{Ik}^*\beta_{Jl}^*\langle \phi_k, \phi_l \rangle + \beta_{Ik}^*\alpha_{Jl}^*\langle \phi_k^*, \phi_l^* \rangle$$
$$= \sum_k \sum_l \delta_{kl}(\alpha_{Ik}^*\beta_{Jl}^* - \beta_{Ik}^*\alpha_{Jl}^*)$$
$$= \sum_k \alpha_{Ik}^*\beta_{Jk}^* - \beta_{Ik}^*\alpha_{Jk}^*$$
$$= (\alpha^*\beta^\dagger - \beta^*\alpha^\dagger)_{IJ} = 0 \quad \Rightarrow \alpha\beta^T - \beta\alpha^T = 1$$

$$[a_I', a_J'^\dagger] = \sum_k \sum_l [\alpha_{Ik}^* a_k - \beta_{Ik}^* a_k^\dagger, -\beta_{Jl} a_l + \alpha_{Jl} a_l^\dagger]$$
$$= \sum_k \sum_l \alpha_{Ik}^*\alpha_{Jl}[a_k, a_l^\dagger] + \beta_{Ik}^*\beta_{Jl}[a_k^\dagger, a_l]$$
$$= \sum_k \alpha_{Ik}^*\alpha_{Jk} - \beta_{Ik}^*\beta_{Jk}$$
$$= (\alpha^*\alpha^T - \beta^*\beta^T)_{IJ} = \delta_{IJ} \quad \Rightarrow \alpha^*\alpha^T - \beta^*\beta^T = 1$$

$$[a_I', a_J'] = \sum_k \sum_l [\alpha_{Ik}^* a_k - \beta_{Ik}^* a_k^\dagger, \alpha_{Jl}^* a_l - \beta_{Jl}^* a_l^\dagger]$$
$$= \sum_k \sum_l -\alpha_{Ik}^*\beta_{Jl}^*[a_k, a_l^\dagger] - \beta_{Ik}^*\alpha_{Jl}^*[a_k^\dagger, a_l]$$
$$= \sum_k \beta_{Ik}^*\alpha_{Jk}^* - \alpha_{Ik}^*\beta_{Jk}^*$$
$$= (\beta^*\alpha^\dagger - \alpha^*\beta^\dagger)_{IJ} = 0 \quad \Rightarrow \alpha^*\beta^\dagger - \beta^*\alpha^\dagger = 0$$